NRR-Patterns: Condition-Bounded Comparison of Five Delayed-Commitment Patterns in Stateful LLM Systems

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Abstract

Delayed-commitment recommendations in large language model systems are often discussed too broadly or reused as if they transfer uniformly across providers and task conditions. This paper compares five delayed-commitment patterns under two matched $t = 0.0$ runs with live provider APIs, and then adds provider-separated stability, temperature robustness, and selected Stage B boundary checks to show where the comparison weakens, reverses, or becomes non-effective. On the matched $t = 0.0$ comparison, Hierarchical Resolution, Delta Update, Lazy Expansion, Conditional Refinement, and Branch Preservation are all positive relative to their own paired baselines at the provider- balanced aggregate level, while Branch remains only weakly positive. The provider-separated view is less uniform: Branch already contains negative Anthropic members before failing at Stage2, and Conditional weakens at Stage2 while retaining a positive provider-balanced aggregate despite an Anthropic-side reversal. The Stage2 robustness result shows that Branch is the clearest fragile case, but the failure shape is tail-risk-dominated rather than provider-wide: the Gemini-side negative mean is driven mainly by one catastrophic row, while the remaining Gemini rows are near-neutral in aggregate. A compact compression comparison over a retained three-pattern subset shows that the observed gains for that tested subset are not reducible to generic prompt shortening: principle-only remains clearly positive within each tested family-relative contrast, whereas compression-only weakens or fails for key cases. Selected Stage B checks then show that a single aggregate comparison view can hide provider-sensitive reversals and non-effective cells. The result is a condition-bounded comparison whose main conclusions are kept honest by explicit provider-separated stability, robustness, sign-flip, and non-effective reporting rather than by a same-scale cross-family ranking or a universal deployment rule.

Keywords: Non-Resolution Reasoning, delayed commitment, stateful LLM systems, paired comparison, provider-separated stability, boundary honesty

1 Introduction

Delayed commitment is useful in many stateful LLM workflows, but recommendations built from that idea are easy to overgeneralize. A pattern that reduces cost in one setting may weaken, reverse, or become non-effective under another provider or operating condition. The

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main problem for this paper is therefore not whether delayed commitment can ever help, but how to compare five tested delayed-commitment patterns on one auditable matched-run design while keeping the main comparison separate from the additional checks used to expose sign-flip and non-effective regions. In this paper, the principal comparison comes from the two matched $t = 0.0$ runs, while the selected Stage B evidence serves as a separate boundary check rather than as part of the main comparison itself.

Within the NRR series, fixed public lines establish first principles and text-to-state mapping [1, 2]. Additional IME, Transfer, and Coupled lines are cited for context rather than as direct evidentiary supports for the claims made here [3, 4, 5]. The role of this paper is narrower and more practical: compare five delayed-commitment patterns under one matched paired design, then report the resulting evidence through provider-separated stability, temperature robustness, and selected Stage B boundary checks.

NRR is not an anti-LLM framework. NRR does not replace standard LLM use. NRR optimizes when to commit and when to defer, under explicit conditions.

Series Consistency Statement: Across the NRR series, unsupported interpretations are not injected at a fixed evidence state, hypothesis-set expansion is evidence-gated, and unobserved possibilities remain explicitly unresolved. In NRR-Patterns, this policy is operationalized by keeping the main five-pattern comparison separate from additional boundary checks and by reporting unstable, sign-flip, and non-effective regions explicitly rather than hiding them behind winner-only summaries.

Series Extensibility Statement: NRR should be read here as a derivation-and-evaluation framework, not as a closed recipe. The five delayed-commitment patterns in NRR-Patterns are an evaluated set under the present matched paired comparison plus explicit robustness and boundary checks, not an exhaustive set of admissible patterns. Additional patterns or operator-family extensions are in scope only when they are tested under the same paired-unit discipline and reported with equally explicit condition-bounded boundary language.

NRR-Patterns compares five delayed-commitment patterns under one matched paired design and then adds provider-separated stability, temperature robustness, and selected Stage B boundary checks to show where that map weakens, reverses, or becomes non-effective.

The five canonical patterns are:

1. Hierarchical Resolution
2. Lazy Expansion
3. Branch Preservation
4. Delta Update
5. Conditional Refinement

These are treated here as delayed-commitment *patterns*, not as the default meaning of lower-level *operator* language used elsewhere in the series. That distinction matters because other papers in the series path also use $(\sigma/\delta, \text{target})$-style operator language for interface-level update primitives, but those mentions are contextual here rather than evidentiary supports for the present pattern-comparison claims [2, 4, 5]. The current paper instead evaluates pattern-level delayed-commitment designs.

1.1 Contributions

1. A fixed matched-run comparison of five delayed-commitment patterns on two matched $t = 0.0$ runs.
2. A separated provider-stability and temperature-robustness read showing where the comparison headline preserves, weakens, or breaks under provider-local condition shifts.

3. A condition-bounded pattern map whose positive and unstable regions are supported by the matched comparison rather than by a same-scale cross-family ranking.

4. Selected Stage B boundary checks for sign-flip and non-effective regions, showing where aggregate comparison reads can hide reversals or weak cells.

5. A compression control over a retained three-pattern subset showing that those tested gains are not reducible to generic prompt shortening.

2 Related Work

Delayed commitment and stateful interaction. Prior NRR work covers the failure mode and text-to-state mapping needed for stateful execution on stateless APIs in fixed public lines [1, 2]. Additional IME, Transfer, and Coupled references are retained here only to mark the broader series path and not as fixed evidentiary supports for the present comparison claims [3, 4, 5]. The present paper does not reopen those layers. Instead, it compares multiple delayed-commitment patterns built on top of them and asks how that comparison should be read under explicit provider-sensitive boundaries.

Efficiency and compression. Context-efficiency work, including retrieval compression and prompt-side compaction, targets token cost and context footprint directly [6, 7]. This paper overlaps in efficiency interest but differs in object: the main question is not simply whether prompts can be shortened, but whether delaying specific commitments yields a usable pattern map that remains condition-bounded once provider-sensitive boundary checks are surfaced. That is why the compression comparison is kept as a control rather than as the paper's main identity.

Boundary-aware evaluation. Evaluation-framework work such as HELM, MT-Bench, and AlpacaEval shows that reported quality or preference conclusions can shift with benchmark protocol, judge setup, and reporting frame [8, 9, 10].

Prompting and calibration sensitivity. Prompting and calibration work likewise shows that effect direction can change with reasoning prompts, aggregation strategy, and label-bias controls [11, 12, 13]. This manuscript adopts the shared lesson in a narrower form: it keeps one exact-replication paired spine, then reports a separated temperature-robustness member plus selected provider-sensitive boundary evidence so the comparison map is not over-read as a same-scale cross-family ranking.

3 Pattern Set and Evaluation Logic

3.1 Pattern Set and Paired Baselines

Each pattern is evaluated under paired baseline-vs-pattern comparison with fixed scenario definitions and prompt templates inherited from the released comparison surfaces. Table 1 states the pattern set, its baseline contrast, and the supported comparison/boundary profile under the current evidence.

3.2 Primary and Supporting Metrics

The headline comparison metric is *provider-balanced weighted reduction* on the exact-replication comparison. The comparison contract for this quantity is recorded in `repro/nrr_patterns_split_comparison_contract_v13_2026-04-12.md`. For a given pattern and run, each provider first contributes its own weighted reduction percentage, $100 \times \sum_i (b_i - p_i) / \sum_i b_i$, over the paired rows in that run's robust source file, where b_i and p_i denote the baseline and pattern token totals for the same paired row. The run-level quantity `weighted_percent_provider_balanced` is then the simple arithmetic mean of those provider-level weighted percentages across

Pattern	Paired baseline contrast	Delayed-commitment axis	Supported comparison/boundary profile
Hierarchical Resolution	flat full-option selection vs category-then-tool selection	decision granularity	positive family-relative benchmark with large within-family reduction
Lazy Expansion	eager full retrieval vs summary-first selective retrieval	content expansion	positive family-relative benchmark with later boundary sensitivity
Branch Preservation	no branch memory vs checkpointed branch restoration	branch destruction	provider-local instability case on the comparison surface
Delta Update	full-state resend vs changed-item-only updates	state retransmission	positive family-relative benchmark
Conditional Refinement	detailed-by-default vs coarse-first refinement	output detail	positive family-relative benchmark with sparse boundary support

Table 1: Canonical pattern catalog for NRR-Patterns. The table fixes the five compared patterns, their paired baseline contrasts, and the evidence profile used to interpret each family under the current comparison-plus-boundary read.

providers. Table 2 reports the two exact-replication run values, their mean, and the separate Stage2 temperature-robustness result, while Table 3 reports the compact provider-separated stability summary derived from the same paired rows. Supporting comparison metrics remain weighted global reduction, median reduction, trimmed reduction, and cost-per-success delta.

3.3 Boundary-Honesty Metrics

The selected boundary metrics are misinterpretation improvement and rework improvement. Here "Stage B" means the historical all-pattern confirmatory boundary-resolution suite in the 'nrr-boundary' package, from which this paper uses only a small subset. The selection rule is recorded in `repro/nrr_patterns_stageb_selection_rule_v2_2026-04-12.md`. The retained pair is the smallest downstream quality/cost pair that can keep the five-pattern comparison honest without turning NRR-Patterns into a second boundary-focused paper. Additional diagnostics such as trace completeness, update consistency, internal coherence error rate, self-contradiction rate, and selected stability diagnostics remain available as secondary surfaces when they materially sharpen the interpretation.

3.4 Reporting Rules and Validity Limits

Use provider-balanced results for the headline comparison and provider-separated results for boundary honesty where aggregation could hide reversals.

The canonical result-language ladder is:

- positive region
- unstable region
- sign-flip region
- non-effective region

The main comparison surface primarily supports the positive/unstable read. Sign-flip and non-effective labels are carried by selected boundary checks where those regions are directly

observed. Accordingly, these labels should not be misread as if they were all outputs of the headline comparison metric alone.

4 Experimental Surfaces

4.1 Main Comparison Surface

The primary experimental surface uses the exact-replication five-pattern comparison in the released NRR-Patterns materials. Here, exact replication means the fixed auditable matched-run outputs named below, not a rerun-capable environment. The provider/model scope is Anthropic Claude Sonnet 4 (`claude-sonnet-4-20250514`), Gemini 2.0 Flash (`gemini-2.0-flash`), and OpenAI GPT-4o mini (`gpt-4o-mini-2024-07-18`):

- Stage1 main ($t = 0.0$)
- Stage1 rep2 ($t = 0.0$)

These two $t = 0.0$ runs define the main exact-replication comparison. The comparison names are recorded in `repro/nrr_patterns_comparison_claims_manifest_v7_2026-04-12.csv`, and the fairness/aggregation contract for separating exact replication from robustness is recorded in `repro/nrr_patterns_split_comparison_contract_v13_2026-04-12.md`. The comparison headline summary is `results/analysis/nrr_patterns_comparison_headline_summary_v3_2026-04-12.csv`, the compact provider-separated stability summary is `results/analysis/nrr_patterns_provider_separated_stability_summary_v3_2026-04-12.csv`, and the headline source members are:

- `results/nrr_patterns_results_allproviders_clean.zip::results/analysis/nrr_patterns_allproviders_pattern_summary_robust_t00.csv`
- `results/nrr_patterns_results_stage1_t00_rep2.zip::results/analysis/nrr_patterns_allproviders_pattern_summary_robust_t00.csv`

The separate temperature-robustness result is:

- `results/nrr_patterns_results_stage2_t03.zip::results/analysis/nrr_patterns_allproviders_pattern_summary_robust_t03.csv`

The paper therefore reports one exact-replication summary while keeping the robustness result separate rather than collapsing all three members into one undifferentiated mean.

4.2 Compression Comparison

The compression comparison remains in the main manuscript as a compact interpretive control. Its role is to test whether the observed gains can be explained by generic prompt compression rather than by delayed commitment itself. The comparison is recorded in `repro/nrr_patterns_comparison_claims_manifest_v7_2026-04-12.csv`. The aggregated control surface is `results/analysis/nrr_patterns_comparison_summary_r2_aggregated.csv`, with per-provider drill-down in `results/analysis/nrr_patterns_comparison_summary_r2_by_provider.csv` and raw comparison traces inside the three provider ZIP artifacts.

4.3 Selected Stage B Boundary-Resolution Surface

Selected Stage B evidence enters this paper as a boundary-check layer, not as the paper's primary confirmatory center. Here, "Stage B" refers to the historical 'nrr-boundary' confirmatory boundary-resolution suite over the same five delayed-commitment families, but this

Pattern	Rows	S1 main	S1 rep2	Rep. mean	S2
Hierarchical	45	+56.752%	+57.606%	+57.179%	+55.737%
Delta	18	+39.074%	+38.956%	+39.015%	+38.980%
Lazy	45	+37.951%	+37.459%	+37.705%	+38.066%
Conditional	18	+37.011%	+35.353%	+36.182%	+32.235%
Branch	45	+2.452%	+3.081%	+2.766%	−21.713%

Table 2: Provider-balanced weighted reduction under the split comparison contract. The headline mean uses only the two exact-replication $t = 0.0$ runs, while Stage2 ($t = 0.3$) is reported separately as the robustness member. The surface is not evidence-symmetric across patterns: Hierarchical, Lazy, and Branch carry $n_{\text{all}} = 45$ paired rows per run, whereas Delta and Conditional carry $n_{\text{all}} = 18$. The magnitudes are not directly comparable across families as a same-scale ranking because each family uses its own paired baseline contrast and token-cost floor.

paper uses only the selected subset named here. The relevant file names are recorded in `repro/nrr_patterns_boundary_claims_manifest_v2_2026-04-12.csv`, and the selection rule is recorded in `repro/nrr_patterns_stageb_selection_rule_v2_2026-04-12.md`. The retained pair `misinterpretation_rate` plus `rework_cost_tokens` is the smallest downstream quality/cost pair needed to keep the comparison honest. In this package, the selected Stage B evidence is traceable to:

- `nrr-boundary/stats/stageb_all/mst_provider_balanced.csv`
- `nrr-boundary/stats/stageb_all/mst_provider_separated.csv`
- `nrr-boundary/stats/stageb_all/mst_sign_flip_boundaries.csv`
- `nrr-boundary/stats/stageb_all/stageb_confirmatory_tests_holm30.csv`

These files are used only to show where the comparison map weakens, reverses, or becomes non-effective. They do not replace the token-efficiency headline, and they do not turn this paper into a second boundary-focused manuscript.

4.4 Secondary Appendix Materials

Ordered-combination follow-up, rep1-vs-rep3 direction details, and no-position ablation are not part of the default main-text claim hierarchy. Ordered composition remains secondary because it introduces a separate provider-by-order interaction story that can easily dominate the paper if over-promoted. The no-position ablation remains useful as a robustness check, but it is not required for the main comparison claim hierarchy. Both therefore remain in the appendix by default.

5 Results: Pattern Comparison

5.1 Provider-Balanced Pattern Map

Across the two exact-replication comparison runs, provider-balanced weighted reduction remains strongly heterogeneous rather than uniform. Table 2 gives the current summary from the comparison headline summary named in Sec. 4, while keeping the Stage2 temperature-robustness member separate from the headline mean.

The resulting exact-replication read should therefore be interpreted with explicit paired-row-volume asymmetry, non-comparable family baseline contrasts, and provider-separated stability in view:

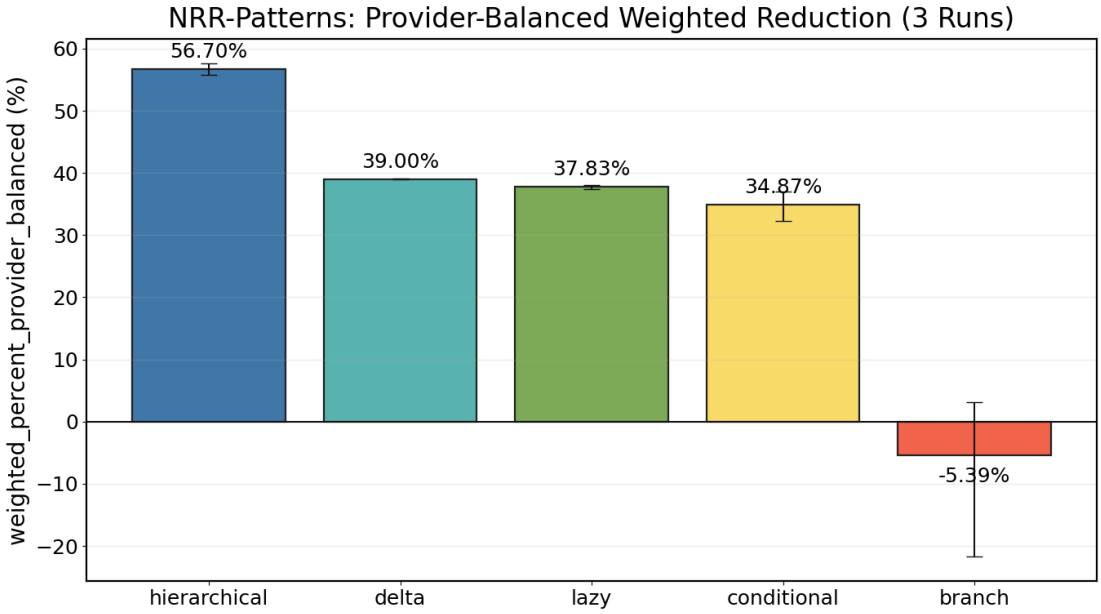

Figure 1: Provider-balanced weighted reduction across the two exact-replication runs plus the separated Stage2 robustness member. Error bars show the observed range over these members for each pattern. The plotted magnitudes remain family-relative against different paired baselines and should not be read as a same-scale cross-family ranking.

- Hierarchical, Delta, and Lazy remain positive for Anthropic, Gemini, and OpenAI across the two exact-replication runs and the separated robustness member.
- Conditional is positive for all three providers on exact replication, but the Stage2 robustness member contains an Anthropic-side reversal even though the provider-balanced aggregate stays positive.
- Branch is aggregate-positive on exact replication but already contains negative Anthropic exact-replication members, then fails the separated robustness member through a Gemini-led, tail-risk-dominated collapse.

5.2 Replication and Temperature Robustness

NRR-Patterns retains the Stage2 member because the main issue is not just exact replication but also whether the same paired comparison object survives a controlled condition shift. That robustness member must be read separately rather than folded into the headline mean. Hierarchical, Delta, and Lazy remain positive for all three providers in both exact-replication runs and in Stage2. Conditional, by contrast, weakens at $t = 0.3$ and flips negative on Anthropic even though the provider-balanced Stage2 aggregate remains positive. Branch is the clearest failure case: it is slightly aggregate-positive in both $t = 0.0$ runs despite negative Anthropic exact-replication members, then turns sharply negative in Stage2 because Gemini contains a tail-risk event that pulls down the provider-balanced aggregate rather than producing a uniform all-provider collapse. That is why the NRR-Patterns reads Branch as a fragile provider-local case rather than as a robust fifth positive family benchmark.

5.3 Provider-Separated Stability Read

The split comparison surface already shows that exact-replication positivity is not the same as universal safety. Table 3 makes that point visible inside the main text rather than leaving it only

Pattern	S1 main (A/G/O)	S1 rep2 (A/G/O)	S2 (A/G/O)	Comparison-side read
Hierarchical	A +41.3 / G +71.0 / O +57.9	A +41.3 / G +74.2 / O +57.3	A +41.7 / G +68.2 / O +57.4	all-provider positive across the exact-replication spine and the separated robustness member
Delta	A +33.1 / G +35.9 / O +48.2	A +33.7 / G +34.6 / O +48.6	A +32.5 / G +38.9 / O +45.6	all-provider positive across the comparison surface, but with lower support volume than Hierarchi-cal/Lazy/Branch
Lazy	A +47.6 / G +41.1 / O +25.1	A +47.6 / G +40.7 / O +24.1	A +47.9 / G +42.1 / O +24.2	all-provider positive on comparison and robustness, with later degradation appearing in the boundary layer
Conditional	A +6.8 / G +64.1 / O +40.1	A +6.3 / G +61.0 / O +38.8	A -4.1 / G +58.0 / O +42.8	exact-replication all-provider positive, but Stage2 contains an Anthropic-side reversal inside an aggregate-positive read
Branch	A -0.1 / G +6.2 / O +1.2	A -1.1 / G +6.4 / O +4.0	A +5.3 / G -74.3 / O +3.9	aggregate-positive exact replication with Anthropic instability, then Gemini-led Stage2 failure

Table 3: Compact provider-separated weighted reduction summary, derived from the provider-separated stability summary artifact in the shipped audit files. Provider order is Anthropic/Gemini/OpenAI. This table makes the comparison-side instability visible in the main text instead of leaving it only inside the shipped paired-delta ZIP members.

in the shipped ZIP members. Branch is the clearest example: it stays slightly aggregate-positive on the exact-replication spine, but Anthropic is already negative in both exact-replication runs, and the Stage2 negative turn is driven by Gemini rather than by uniform all-provider collapse. Conditional is less severe but still illustrates the same principle by weakening materially at $t = 0.3$ while carrying an Anthropic-side reversal inside an aggregate-positive provider-balanced read. The other three patterns are substantially more stable across the split comparison read, but even there the correct interpretation is still condition-bounded rather than universal. This is why NRR-Patterns stops at a comparison-led family benchmark map and then carries boundary honesty separately, rather than converting the comparison table into a deployment-wide ranking.

The Branch failure shape is also narrower than a broad provider-wide collapse. In the Stage2 Gemini paired-delta member `results/nrr_patterns_results_stage2_t03.zip::results/analysis/nrr_patterns_prod_gemini_clean_t03_paired_deltas.csv`, the worst Branch row is `scenario3 / repetition3`, where the token count changes from `3372` to `38998` (-1056.524318%). The Gemini Branch provider-weighted Stage2 value is -74.326453% with that row included, but the compact provider-separated summary also records that it would be only -0.530303% without that one worst row. The appropriate reading is therefore tail-risk-dominated robustness failure, not a broad all-row or all-provider collapse.

Pattern	Rows	Split status	Boundary note
Hierarchical	45	robust positive region	remains strong on the exact-replication spine, but later boundary views still matter for provider-sensitive weakening cells
Lazy	45	robust positive region	exact-replication positive and robustness-positive, but carried-forward boundary views expose weaker and degrading cells
Branch	45	aggregate-positive instability / tail-risk robustness failure	exact replication is only aggregate-positive because Anthropic is already negative, and the separated robustness member fails through a Gemini-led tail-risk event
Delta	18	robust positive region	exact-replication positive and robustness-positive, but selected boundary checks expose weaker cells
Conditional	18	aggregate-positive with Stage2 provider reversal	clearest positive provider-mean support on the retained key metrics, but Stage2 already contains an Anthropic-side comparison reversal and the boundary layer stays sparse/non-uniform

Table 4: NRR-Patterns status ladder under the split comparison contract. Status is assigned from the exact-replication spine plus the separated robustness member, while the rows/run column exposes the unequal paired-row counts behind the five-pattern map. Unequal paired-row volume is not equivalent to unequal independent support thickness.

5.4 Pattern Status Ladder

The end of the comparison section summarizes the main read in one compact status ladder. Table 4 states the current NRR-Patterns summary.

6 Results: Boundary and Control Checks

6.1 Compression Is Not a Drop-In Substitute

The compression control stays in the main text because it answers the most natural skepticism against a comparison-led delayed-commitment paper: perhaps the gains come only from shorter prompts. The current aggregated comparison, however, does not support that simpler story for the retained three-pattern subset. Table 5 shows that principle-only remains clearly positive for all three tested patterns, whereas compression-only is negative for Hierarchical and Delta and only near-neutral for Lazy.

The comparative structure matters. Principle-only remains the strongest read for all three tested patterns, while the combination condition stays positive but weaker than principle-only. The provider-separated control rows also show why this section must remain condition-bounded rather than aggregate-only: for Hierarchical, compression-only is negative on Anthropic and OpenAI but positive on Gemini, while for Lazy it is positive on Anthropic and OpenAI but strongly negative on Gemini. Compression-only therefore functions as a useful control for this retained subset, but not as an equivalent or provider-invariant substitute for delayed-commitment

Pattern	Principle-only	Compression-only	Principle+compression
Delta	+38.148%	−25.623%	+3.324%
Hierarchical	+57.212%	−8.224%	+36.675%
Lazy	+38.207%	+1.523%	+8.589%

Table 5: Provider-balanced weighted reduction for the retained r2 compression comparison. Values are read from the aggregated r2 comparison summary artifact described in Sec. 4.

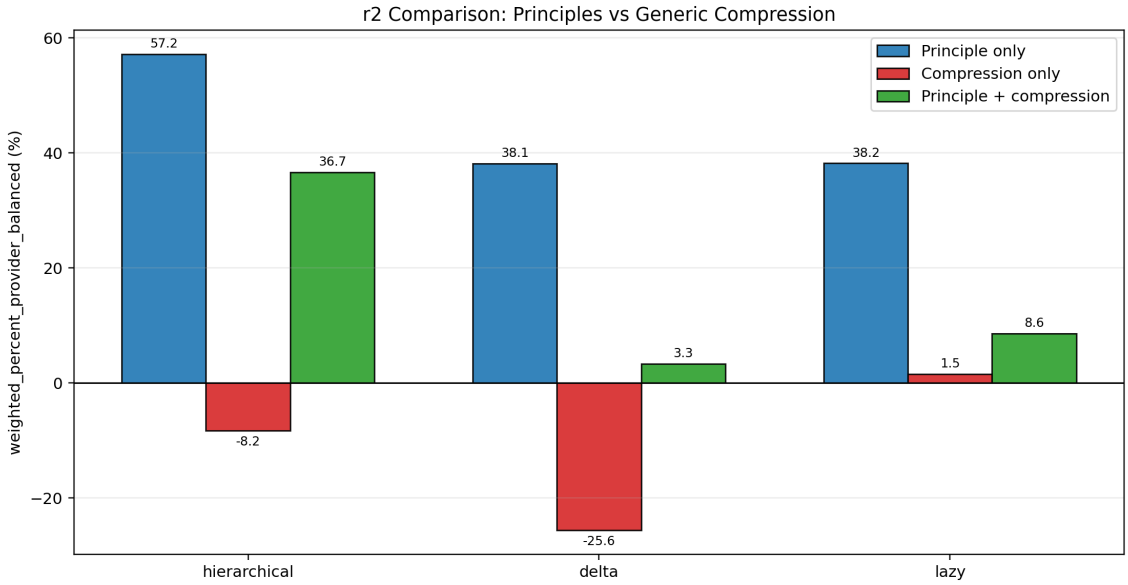

Figure 2: Compression control for the retained three-pattern subset. Principle-only remains strongest, while compression-only weakens or fails for key cases.

patterning on that tested three-pattern slice.

6.2 Selected Stage B Boundary Checks

Selected Stage B evidence is included only to show where provider-separated quality behavior weakens, reverses, or becomes non-effective. This Stage B layer is a selected subset of the historical 'nrr-boundary' confirmatory suite, not a new manuscript-native experiment block. The retained `misinterpretation_rate` plus `rework_cost_tokens` pair is the smallest downstream quality/cost pair needed for honest reporting. Table 6 summarizes the two key metrics retained for this role.

These boundary checks change how the comparison map should be read:

- Conditional has the clearest positive provider-mean support on the retained Stage B key metrics, but that support remains sparse rather than uniformly active.
- Lazy is the clearest degraded pattern on the retained confirmatory surface, with negative provider-balanced means on both retained key metrics.
- Hierarchical, Delta, and Branch show provider-sensitive weakening or reversal cells rather than a clean all-provider positive read.
- Sign-flip and non-effective regions must be reported explicitly rather than hidden behind pooled summaries.

Pattern	Misinterpretation improvement	Rework improvement (tokens)
Conditional	+0.0781	+122.6481
Hierarchical	+0.0120	−12.1852
Delta	+0.0028	−26.6296
Branch	−0.0006	−24.7222
Lazy	−0.1242	−234.3704

Table 6: Selected Stage B provider-balanced key-metric improvements, read from the Stage B provider-balanced artifact described in Sec. 4. Positive values indicate improvement over baseline.

Pattern	All-provider zero metrics on selected Stage B
Conditional	internal coherence error rate; trace completeness
Delta	internal coherence error rate; trace completeness
Branch	internal coherence error rate; trace completeness
Hierarchical	none in the selected all-provider-zero slice
Lazy	none in the selected all-provider-zero slice

Table 7: Selected Stage B all-provider non-effective cells. A cell appears here when the provider-separated honesty artifact reports zero-direction behavior for Anthropic, Gemini, and OpenAI under the current setup. Provider-specific zero or near-zero cells remain in the shipped provider-separated artifact even when they do not rise to the all-provider slice shown here.

The provider-separated read makes the same point more sharply. Conditional remains positive on provider means for misinterpretation and rework across Anthropic, OpenAI, and Gemini, but the support is still sparse rather than uniformly active: in `nrr-boundary/stats/stageb_all/stageb_confirmatory_tests_holm30.csv`, misinterpretation carries $n_{\text{nonzero}} = 25/54$ and rework carries 33/54, and the provider-separated rows still contain substantial zero-count mass plus some negative rework cases. By contrast, Hierarchical flips negative on Gemini for both retained key metrics, Delta is negative on Anthropic and Gemini for rework, Branch is negative on Gemini for both retained key metrics, and Lazy is strongly negative on Gemini while only weakly positive or near-neutral elsewhere. On the comparison surface, this same provider-local structure is why Branch should be read as a localized instability case rather than as a uniformly collapsing family. Stage B therefore functions as a boundary-check layer rather than as a replacement for the main comparison.

Sign-flip evidence is also direct rather than rhetorical. In the current Stage B boundary artifact, sign-flip boundaries appear in 20 of 45 pattern-metric cells. Conditional shows no Stage B sign-flip cells under this setup, while Branch, Delta, Hierarchical, and Lazy all show sign-flip boundaries on multiple metrics. Figure 3 is included here for compact honesty reporting rather than for boundary-paper completeness.

The confirmatory results are consistent with this reading. In `nrr-boundary/stats/stageb_all/stageb_confirmatory_tests_holm30.csv`, only five Holm-significant cells survive correction: two positive cells for Conditional (misinterpretation and rework) and three degraded cells for Lazy (update consistency, internal coherence, and self-contradiction). This is enough to keep the positive comparison story honest, but not enough to turn the paper into a second confirmatory study of operator boundaries.

Non-effective reporting is also part of the paper rather than a supplement-only afterthought. Table 7 lists the selected Stage B metrics whose provider-separated artifact shows all-provider zero cells under the current setup. These cells do not overturn the main comparison read by themselves, but they do mark regions where a positive interpretation should stop short of claiming an active effect.

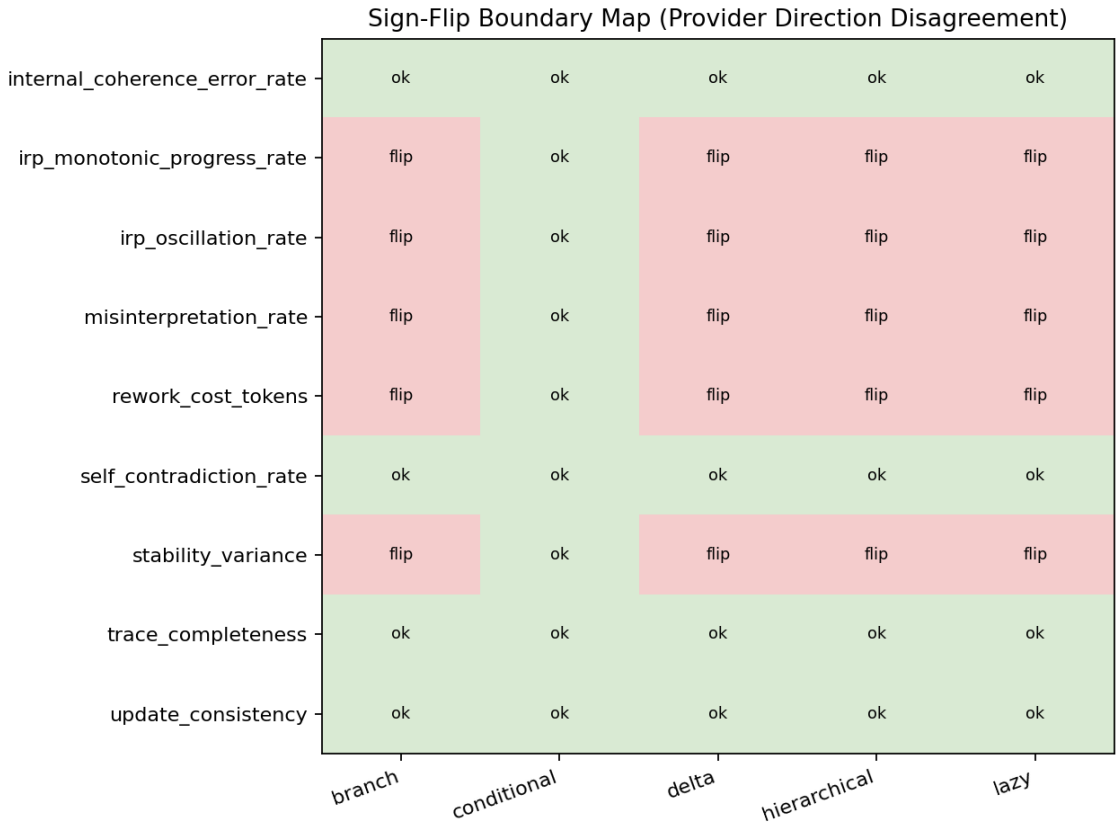

Figure 3: Selected Stage B sign-flip boundary map carried forward as the honesty layer for NRR-Patterns.

7 Discussion

7.1 What the Pattern Map Supports

The supported read is condition-bounded utility, not universal superiority. On the exact-replication comparison, all five patterns are aggregate-positive relative to their own paired baselines, but they do not remain equally strong once the provider-separated stability table, the robustness result, and the boundary evidence are added. Hierarchical and Delta remain robustly positive on the split comparison surface. Lazy remains positive on the exact-replication comparison and the robustness result, but its boundary read is degraded rather than robustly positive. Conditional stays aggregate-positive but already contains a Stage2 provider-side reversal. Branch fails even earlier by showing exact-replication Anthropic instability and then losing positive status on the robustness result through a tail-risk-dominated provider-local failure. The appropriate output is therefore a family-benchmark pattern map, not a same- scale ranking.

7.2 Why Provider Sensitivity Changes the Read

Provider-balanced summaries are useful for the exact-replication headline, but the robustness result and provider-separated views are necessary where aggregation can hide reversals or tail-risk concentration. This paper therefore distinguishes four evidence roles: exact replication for the main family-benchmark read, provider-separated stability for comparison-side reversals and concentration, robustness for controlled condition-shift stress, and boundary checks for the caveat layer. Collapsing those roles would either overstate the comparison or overinflate the boundary section.

7.3 Why Compression Is Not Enough

The compact control against generic prompt compression matters because it distinguishes delayed-commitment patterning from a simpler prompt shortening story. On the retained r2 surface, principle-only stays clearly positive where compression-only weakens or fails for the three tested patterns. This does not prove that compression is never useful; it shows that the tested subset gains are not adequately explained by prompt shortening alone.

7.4 How to Read Unstable, Sign-Flip, and Non-Effective Regions

These regions are not embarrassing leftovers. They are part of the contribution because they define where reuse becomes unsafe, unsupported, or too weak to generalize under the current evidence state. In this paper, unstable is primarily a comparison-side label, while sign-flip and non-effective are boundary-honesty labels. Keeping those roles distinct is part of the manuscript's scientific honesty.

7.5 Series Position

NRR-Patterns is the pattern-comparison layer upstream of Energy. It is not the calibration paper, and it is not the downstream closure paper. Its job is to make the comparison evidence clear enough that later calibration and closure work do not inherit a blurred pattern story.

8 Limitations

This paper cites traceable comparison and Stage B values directly from the released analysis files, and the naming conventions are recorded in the manifests. The released artifact set is auditable rather than a full rerun bundle: it includes the manuscript assets, the comparison and boundary manifests, a checksum manifest, and the named comparison and boundary files needed for inspection. Here, exact replication means that the fixed matched-run outputs can be audited and examined from preserved summaries and run-specific members; it does not mean that the distributed bundle is itself a full independent rerun environment. Provider sensitivity remains real and must not be hidden by a single aggregate summary, and the five-pattern map is based on unequal paired-row volume across patterns. That paired-row volume is useful for auditability, but it is not the same thing as independent support thickness. Ordered composition remains outside the main-text claim hierarchy, and no-position ablation remains appendix-first rather than main-text essential.

9 Conclusion

NRR-Patterns supports a condition-bounded pattern map over five delayed- commitment patterns under a split comparison contract. On the exact-replication comparison, all five patterns remain aggregate-positive relative to their own paired baselines, but the provider-separated stability summary and the temperature-robustness result show that those positives are not equally stable. Hierarchical and Delta remain robustly positive on the split comparison surface, Lazy stays positive on comparison and robustness while degrading in the boundary checks, Conditional remains aggregate-positive but already contains a Stage2 provider-side reversal, and Branch becomes the clearest fragile provider-local case because exact-replication Anthropic instability is followed by a Gemini-led, tail-risk-dominated Stage2 failure rather than by uniform collapse across providers. A compact compression control over the retained three-pattern subset shows that those tested gains are not reducible to generic prompt shortening alone. The supported output is therefore not a universal ranking, but a comparison-led family benchmark map with explicit provider-separated stability, robustness, sign-flip, and non-effective regions.

10 Reproducibility and Appendix Home

The current NRR-Patterns release is organized as an auditable evidence package rather than a full end-to-end rerun bundle. The package map is recorded in `repro/nrr_patterns_package_map_v20_2026-04-12.md`, the split comparison contract is recorded in `repro/nrr_patterns_split_comparison_contract_v13_2026-04-12.md`, the Stage B selection rule is recorded in `repro/nrr_patterns_stageb_selection_rule_v2_2026-04-12.md`, the release-manifest checksum anchor is `repro/nrr_patterns_repro_checksums_sha256.txt`, and the shipped audit checksum manifest is `repro/nrr_patterns_audit_surface_sha256_v20_2026-04-12.txt`. The main comparison files are:

- `results/analysis/nrr_patterns_comparison_headline_summary_v3_2026-04-12.csv`
- `results/analysis/nrr_patterns_provider_separated_stability_summary_v3_2026-04-12.csv`
- `results/nrr_patterns_results_allproviders_clean.zip::results/analysis/nrr_patterns_allproviders_pattern_summary_robust_t00.csv`
- `results/nrr_patterns_results_allproviders_clean.zip::results/analysis/nrr_patterns_allproviders_paired_deltas_merged_t00.csv`
- `results/nrr_patterns_results_stage1_t00_rep2.zip::results/analysis/nrr_patterns_allproviders_pattern_summary_robust_t00.csv`
- `results/nrr_patterns_results_stage2_t03.zip::results/analysis/nrr_patterns_allproviders_pattern_summary_robust_t03.csv`
- `results/nrr_patterns_results_stage2_t03.zip::results/analysis/nrr_patterns_allproviders_paired_deltas_merged_t03.csv`
- `results/nrr_patterns_results_stage2_t03.zip::results/analysis/nrr_patterns_prod_gemini_clean_t03_paired_deltas.csv`
- `results/analysis/nrr_patterns_comparison_summary_r2_aggregated.csv`

The selected boundary-check files are:

- `nrr-boundary/stats/stageb_all/mst_provider_balanced.csv`
- `nrr-boundary/stats/stageb_all/mst_provider_separated.csv`
- `nrr-boundary/stats/stageb_all/mst_sign_flip_boundaries.csv`
- `nrr-boundary/stats/stageb_all/stageb_confirmatory_tests_holm30.csv`

Appendix materials include the no-position ablation (`results/nrr_patterns_no_position_t00.zip`, `results/analysis/nrr_patterns_no_position_summary.csv`). An ordered-combination follow-up is also retained outside the main-text claim hierarchy, but it is named for traceability only and is omitted from the current default audit bundle. Only the no-position ablation is shipped in the current audit bundle.

The current NRR-Patterns checksum manifest is `manuscript/checksums_active_review_surface_sha256.txt`. In the current release, `scripts/verify_current_package.sh` verifies the manuscript asset set together with the release manifests, while `scripts/verify_audit_surface.py` verifies the listed audit files against the audit checksum manifest. Together, the package map, the split comparison contract, the Stage B selection rule, the comparison summaries, the release-manifest checksum anchor, the audit checksum manifest, and the current manuscript checksum manifest define the documented artifact set for this paper.

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